

## SAFETY DATA SHEET

Revision Date 25-Mar-2024

Revision Number 2

### 1. Identification

**Product Name** Zinc oxide, NanoShield ZN-2000, 50% in water, colloidal dispersion with nonionic dispersant

**Cat No. :** 44904

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### Emergency Telephone Number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

#### Label Elements

None required

#### Hazards not otherwise classified (HNOC)

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component              | CAS No    | Weight % |
|------------------------|-----------|----------|
| Zinc oxide             | 1314-13-2 | 50       |
| Water                  | 7732-18-5 | 47       |
| Proprietary dispersant | N/A       | 3        |

#### 4. First-aid measures

|                                            |                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes.                       |
| <b>Inhalation</b>                          | Remove to fresh air.                                                                     |
| <b>Ingestion</b>                           | Do NOT induce vomiting.                                                                  |
| <b>Most important symptoms and effects</b> | No information available.                                                                |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                    |

#### 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | Not applicable           |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

#### Specific Hazards Arising from the Chemical

Do not allow run-off from fire-fighting to enter drains or water courses.

#### Hazardous Combustion Products

None known.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                                                                                                 |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |

**Methods for Containment and Clean Up** No information available.

## 7. Handling and storage

**Handling** Ensure adequate ventilation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component  | ACGIH TLV                                              | OSHA PEL                                                                                                                                                                   | NIOSH                                                                                                                  | Mexico OEL (TWA)                                       |
|------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Zinc oxide | TWA: 2 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) STEL: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup> | IDLH: 500 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup><br>Ceiling: 15 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas.

### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Particulates filter conforming to EN 143. or. Inorganic gases and vapours filter. Type B. Grey. conforming to EN14387.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | dispersion               |
| Appearance                       | No information available |
| Odor                             | No information available |
| Odor Threshold                   | No information available |
| pH                               | Not applicable           |
| Melting Point/Range              | No data available        |
| Boiling Point/Range              | No information available |
| Flash Point                      | Not applicable           |
| Evaporation Rate                 | No information available |
| Flammability (solid,gas)         | No information available |
| Flammability or explosive limits |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |
| Vapor Pressure                   | No information available |
| Vapor Density                    | No information available |
| Specific Gravity                 | No information available |

|                                        |                          |
|----------------------------------------|--------------------------|
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactive Hazard                  | None known, based on information available |
| Stability                        | Stable under normal conditions.            |
| Conditions to Avoid              | Incompatible products.                     |
| Incompatible Materials           | Strong oxidizing agents                    |
| Hazardous Decomposition Products | None under normal use conditions           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.   |
| Hazardous Reactions              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Mist LC50   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |
| Vapor LC50  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

#### Component Information

| Component  | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation           |
|------------|---------------------------|------------------------------|---------------------------|
| Zinc oxide | LD50 > 5000 mg/kg ( Rat ) | LD50 > 2000 mg/kg, 24h (Rat) | LC50 > 5.7 mg/L, 4h (Rat) |
| Water      | -                         | -                            | -                         |

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component              | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------|-----------|------------|------------|------------|------------|------------|
| Zinc oxide             | 1314-13-2 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Water                  | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Proprietary dispersant | N/A       | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

|                          |            |
|--------------------------|------------|
| STOT - single exposure   | None known |
| STOT - repeated exposure | None known |

Aspiration hazard No information available

Symptoms / effects, both acute and delayed No information available

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component  | Freshwater Algae | Freshwater Fish                             | Microtox   | Water Flea |
|------------|------------------|---------------------------------------------|------------|------------|
| Zinc oxide | Not listed       | LC50: = 1.55 mg/L, 96h static (Danio rerio) | Not listed | Not listed |

Persistence and Degradability No information available

Bioaccumulation/ Accumulation No information available.

Mobility No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN3082  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
 Technical Name Zinc oxide  
 Hazard Class 9  
 Packing Group III

### TDG

UN-No UN3082  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
 Hazard Class 9  
 Packing Group III

### IATA

UN-No UN3082  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
 Hazard Class 9  
 Packing Group III

### IMDG/IMO

UN-No UN3082  
 Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.  
 Hazard Class 9  
 Packing Group III

## 15. Regulatory information

### United States of America Inventory

| Component | CAS No | TSCA | TSCA Inventory notification - | TSCA - EPA Regulatory |
|-----------|--------|------|-------------------------------|-----------------------|
|-----------|--------|------|-------------------------------|-----------------------|

|                        |           |   | <b>Active-Inactive</b> | <b>Flags</b> |
|------------------------|-----------|---|------------------------|--------------|
| Zinc oxide             | 1314-13-2 | X | ACTIVE                 | -            |
| Water                  | 7732-18-5 | X | ACTIVE                 | -            |
| Proprietary dispersant | N/A       | - | -                      | -            |

**Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical  
Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

X = listed, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| <b>Component</b>       | <b>CAS No</b> | <b>DSL</b> | <b>NDSL</b> | <b>EINECS</b> | <b>PICCS</b> | <b>ENCS</b> | <b>ISHL</b> | <b>AICS</b> | <b>IECSC</b> | <b>KECL</b> |
|------------------------|---------------|------------|-------------|---------------|--------------|-------------|-------------|-------------|--------------|-------------|
| Zinc oxide             | 1314-13-2     | X          | -           | 215-222-5     | X            | X           | X           | X           | X            | KE-35565    |
| Water                  | 7732-18-5     | X          | -           | 231-791-2     | X            | X           |             | X           | X            | KE-35400    |
| Proprietary dispersant | N/A           | -          | -           | -             | -            | -           |             | -           | -            | -           |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations**

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| <b>Component</b> | <b>CAS No</b> | <b>Weight %</b> | <b>SARA 313 - Threshold<br/>Values %</b> | <b>SARA 313 - Reporting<br/>thresholds</b> |
|------------------|---------------|-----------------|------------------------------------------|--------------------------------------------|
| Zinc oxide       | 1314-13-2     | 50              | 1.0 %                                    | -                                          |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| <b>Component</b> | <b>CWA - Hazardous<br/>Substances</b> | <b>CWA - Reportable<br/>Quantities</b> | <b>CWA - Toxic Pollutants</b> | <b>CWA - Priority Pollutants</b> |
|------------------|---------------------------------------|----------------------------------------|-------------------------------|----------------------------------|
| Zinc oxide       | -                                     | -                                      | X                             | -                                |

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and  
Health Administration

Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

## U.S. State Right-to-Know Regulations

| Component  | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|------------|---------------|------------|--------------|----------|--------------|
| Zinc oxide | X             | X          | X            | -        | X            |
| Water      | -             | -          | X            | -        | -            |

## U.S. Department of Transportation

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

## U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

## Other International Regulations

Mexico - Grade No information available

## Authorisation/Restrictions according to EU REACH

| Component              | CAS No    | REACH (1907/2006) -<br>Annex XIV - Substances<br>Subject to Authorization | REACH (1907/2006) -<br>Annex XVII - Restrictions<br>on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 -<br>Candidate List of<br>Substances of Very High<br>Concern (SVHC) |
|------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Zinc oxide             | 1314-13-2 | -                                                                         | Use restricted. See item<br>75.<br>(see link for restriction<br>details)               | -                                                                                                                 |
| Water                  | 7732-18-5 | -                                                                         | -                                                                                      | -                                                                                                                 |
| Proprietary dispersant | N/A       | -                                                                         | -                                                                                      | -                                                                                                                 |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component              | CAS No    | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|------------------------|-----------|----------------|---------------------------------|------------------------------|--------------------------------------------------|
| Zinc oxide             | 1314-13-2 | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |
| Water                  | 7732-18-5 | Listed         | Not applicable                  | Not applicable               | Not applicable                                   |
| Proprietary dispersant | N/A       | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component  | CAS No    | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|------------|-----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| Zinc oxide | 1314-13-2 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Annex I - Y23                         |

|                        |           |                |                |                |                |
|------------------------|-----------|----------------|----------------|----------------|----------------|
| Water                  | 7732-18-5 | Not applicable | Not applicable | Not applicable | Not applicable |
| Proprietary dispersant | N/A       | Not applicable | Not applicable | Not applicable | Not applicable |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 25-Mar-2024  
**Print Date** 25-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**